



Supermarket Ordering, Invoicing, and Sales

Introduction

Exploratory Data Analysis (EDA) is an important process in data science used to understand datasets, identify patterns, clean data, and extract meaningful business insights. In this project, three different business-related datasets — **Invoices**, **OrderLeads**, and **SalesTeam** — were analyzed using Python libraries such as Pandas, Matplotlib, and Seaborn.

The main objective of this project was to explore customer ordering behavior, sales lead performance, and sales team management using statistical analysis and visualizations. Multiple EDA techniques were applied, including:

- Data cleaning
- Missing value analysis
- Duplicate checking
- Statistical summaries
- Trend analysis
- Correlation analysis
- Data visualization

Through this project, raw business data was transformed into useful insights that can support smarter business decisions and operational improvements.

Dataset Overview

1. Invoices Dataset

The Invoices dataset contains information related to:

- Company names
- Meal prices
- Meal types
- Invoice records
- Order dates

This dataset was analyzed to understand customer ordering behavior, pricing

trends, and company order activity. Different visualizations such as meal price distributions, meal type counts, yearly order trends, and correlation heatmaps were created to identify business patterns.

2. OrderLeads Dataset

The OrderLeads dataset contains:

- Order values
- Conversion status
- Company details
- Sales lead information
- Order dates

This dataset was used to analyze customer conversion performance and sales opportunities. The analysis focused on identifying high-value customers, conversion trends, and business growth patterns through visualizations and statistical analysis.

3. SalesTeam Dataset

The SalesTeam dataset includes:

- Sales representative names
- Company IDs
- Company assignments
- Workforce management information

This dataset was analyzed to evaluate sales team performance and workload distribution. Different charts and visualizations were used to identify top-performing sales representatives, workload imbalances, and company management patterns.

Invoices.csv

```
In [4]: import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
```

```
In [10]: df = pd.read_csv("/content/Invoices.csv")
```

How many rows and columns are in the dataset?

```
In [12]: df.shape
```

```
Out[12]: (50017, 8)
```

What are the column names in the dataset?

```
In [13]: df.columns
```

```
Out[13]: Index(['Order Id', 'Date', 'Meal Id', 'Company Id', 'Date of Meal',  
              'Participants', 'Meal Price', 'Type of Meal'],  
              dtype='object')
```

What do the first 5 records look like?

```
In [11]: df.head()
```

```
Out[11]:
```

	Order Id	Date	Meal Id	Company Id	Date of Meal
0	839FKFW2LLX4LMBB	27-05-2016	INBUX904GIHI8YBD	LJKS5NK6788CYMUU	20:00:00
1	970X39BGVMHODLJM	27-09-2018	J0MMOOPP709DIDIE	LJKS5NK6788CYMUU	20:00:00
2	041ORQM5OIHTIU6L	24-08-2014	E4UJLQNCI16UX5CS	LJKS5NK6788CYMUU	14:00:00
3	YT796QI18WNGZ7ZJ	12-04-2014	C9SDFHF7553BE247	LJKS5NK6788CYMUU	21:00:00
4	6YLROQT27B6HRF4E	28-07-2015	48EQXS6IHYNZDDZ5	LJKS5NK6788CYMUU	14:00:00

What do the last 5 records look like?

```
In [14]: df.tail()
```

Out[14]:

	Order Id	Date	Meal Id	Company Id
50012	4OMS8ZSA0UX8LHWI	20-09-2017	1TD5MROATV1NHZ4Y	E4K99D4JR9E40VE1
50013	RR0VKJN8V0KHNKGG	19-03-2018	22EX9VZSJKHP4AIP	E4K99D4JR9E40VE1
50014	STJ6QJC30WPRM93H	21-09-2017	LMX18PNGWCIMG1QW	E4K99D4JR9E40VE1
50015	QHEUIYNC0XQX7GDR	28-01-2018	4U0VH2TGQL30X23X	E4K99D4JR9E40VE1
50016	NKHFWT5I2J9LPAPG	06-09-2017	ORWFRT5TUSYGNYG7	E4K99D4JR9E40VE1

What are the data types of each column?

In [63]: `df.dtypes`

Out[63]:

	0
Sales Rep	object
Sales Rep Id	object
Company Name	object
Company Id	object

dtype: object

What is the overall structure and data types of the dataset?

In [15]: `df.info()`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50017 entries, 0 to 50016
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Order Id              50017 non-null  object
1   Date                  50017 non-null  object
2   Meal Id               50017 non-null  object
3   Company Id           50017 non-null  object
4   Date of Meal          50017 non-null  object
5   Participants          50017 non-null  object
6   Meal Price            50017 non-null  int64
7   Type of Meal          50017 non-null  object
dtypes: int64(1), object(7)
memory usage: 3.1+ MB

```

What are the statistical summaries of the numerical columns in the dataset?

```
In [17]: df.describe()
```

```
Out[17]:
```

	Meal Price
count	50017.000000
mean	403.256713
std	234.128913
min	6.000000
25%	204.000000
50%	398.000000
75%	593.000000
max	1065.000000

How many duplicate rows are in the dataset?

```
In [16]: df.duplicated()
```

Out[16]:

0	
0	False
1	False
2	False
3	False
4	False
...	...
50012	False
50013	False
50014	False
50015	False
50016	False

50017 rows × 1 columns

dtype: bool

How many missing values are in the dataset?

In [18]: `df.isnull().sum()`

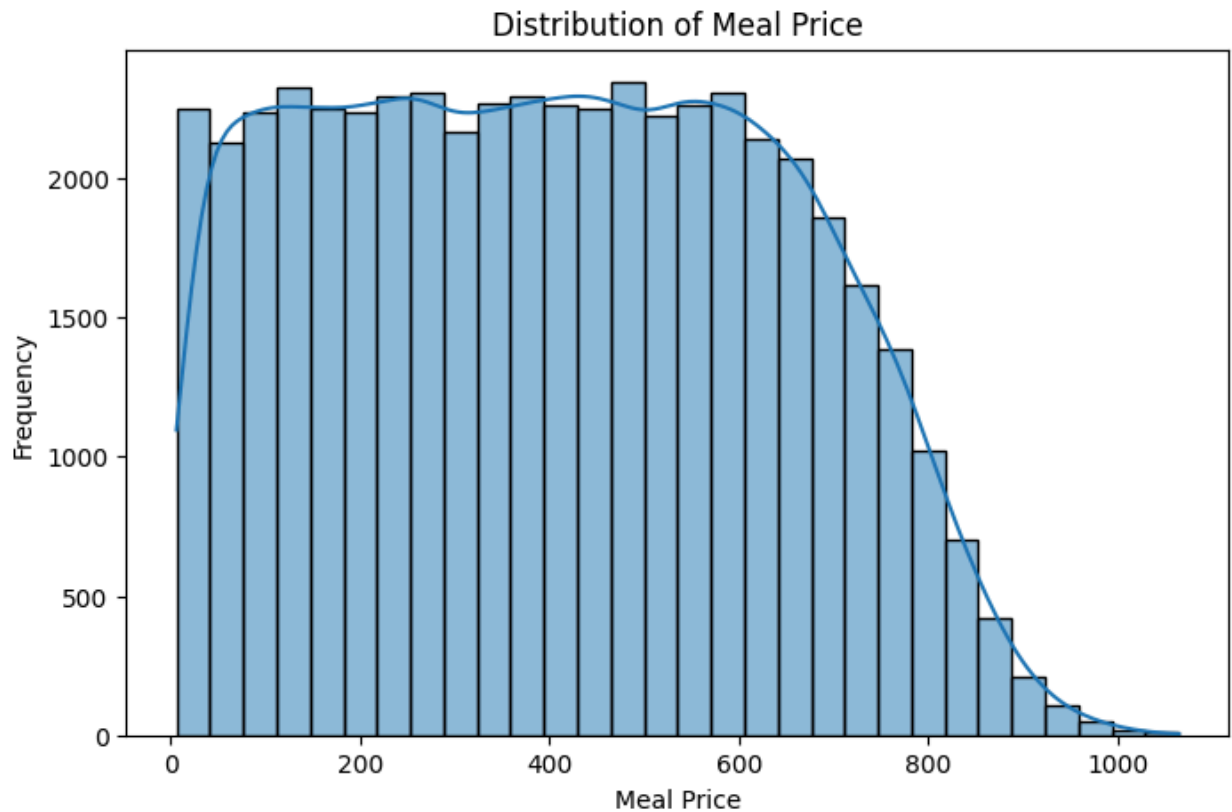
Out[18]:

0	
Order Id	0
Date	0
Meal Id	0
Company Id	0
Date of Meal	0
Participants	0
Meal Price	0
Type of Meal	0

dtype: int64

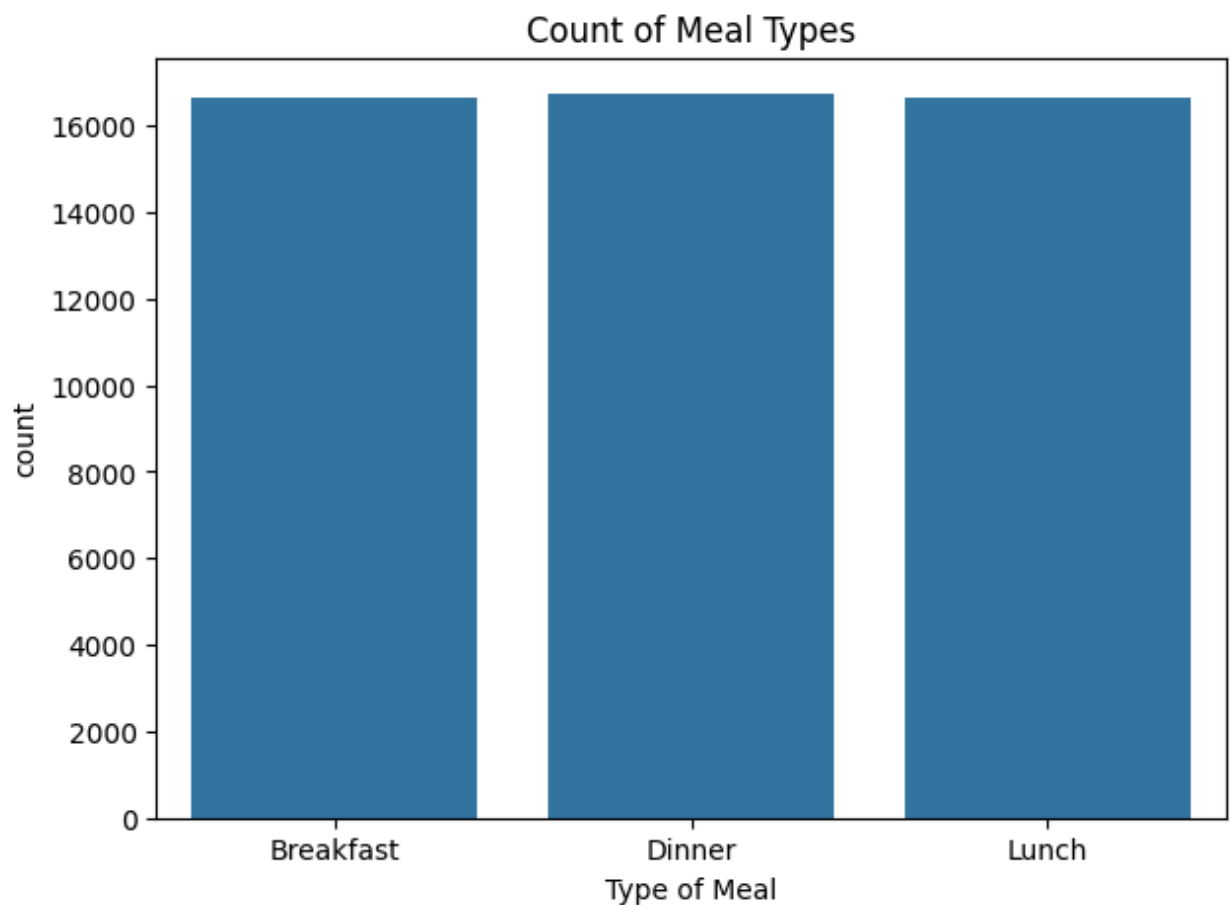
What is the distribution of meal prices?

```
In [21]: plt.figure(figsize=(8,5))
sns.histplot(df['Meal Price'], bins=30, kde=True)
plt.title("Distribution of Meal Price")
plt.xlabel("Meal Price")
plt.ylabel("Frequency")
plt.show()
```



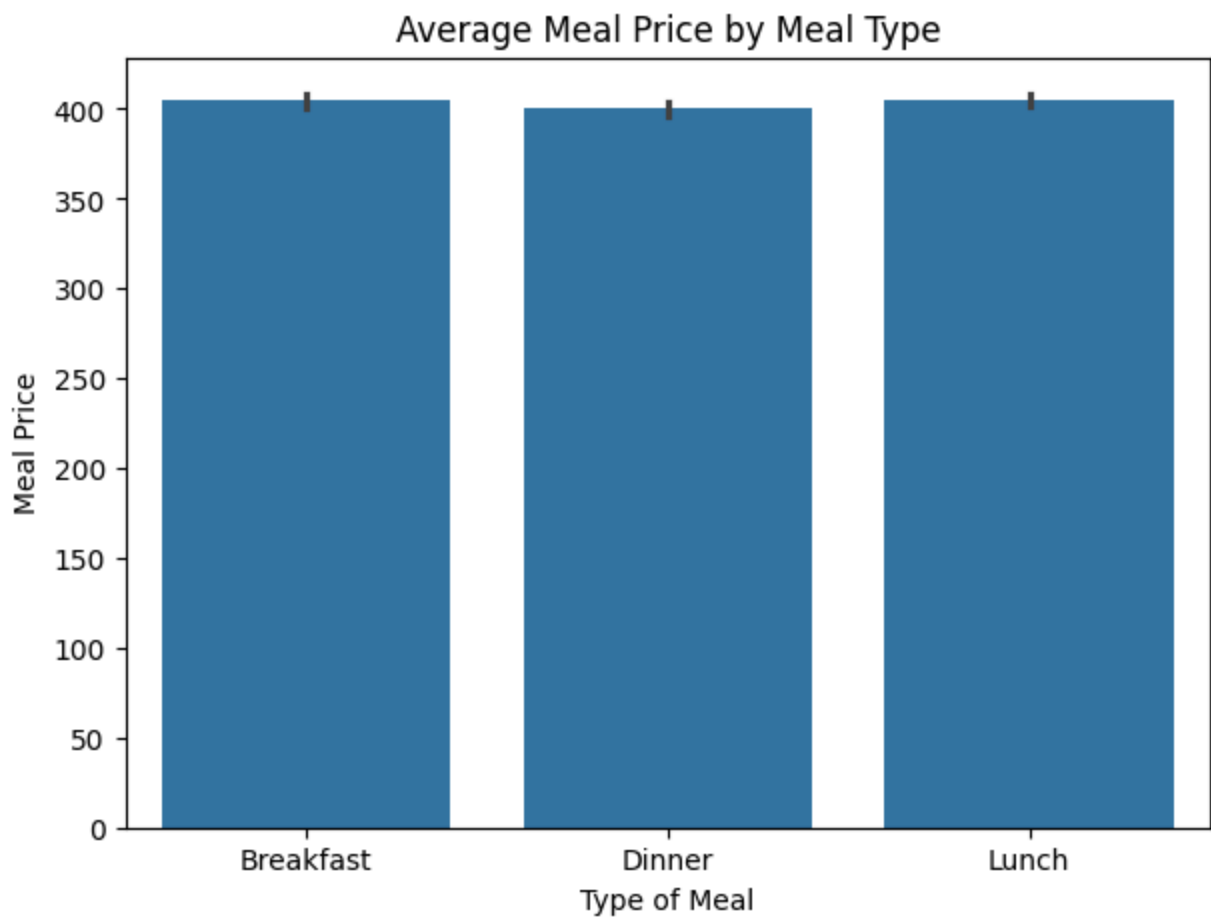
Which meal type has the highest count?

```
In [22]: plt.figure(figsize=(7,5))
sns.countplot(x='Type of Meal', data=df)
plt.title("Count of Meal Types")
plt.show()
```



What is the average meal price for each meal type?

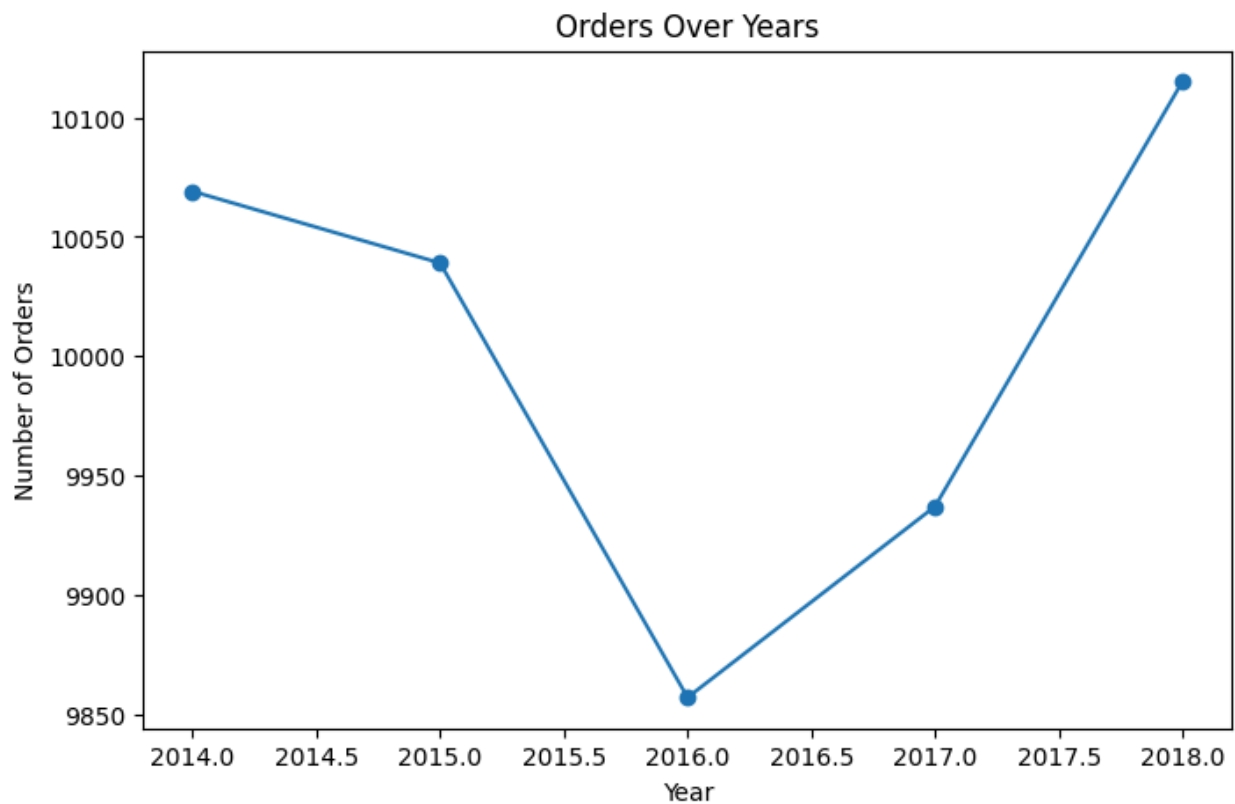
```
In [23]: plt.figure(figsize=(7,5))
sns.barplot(x='Type of Meal', y='Meal Price', data=df)
plt.title("Average Meal Price by Meal Type")
plt.show()
```

How has the number of orders changed over the years?

```
In [24]: df['Year'] = df['Date'].dt.year

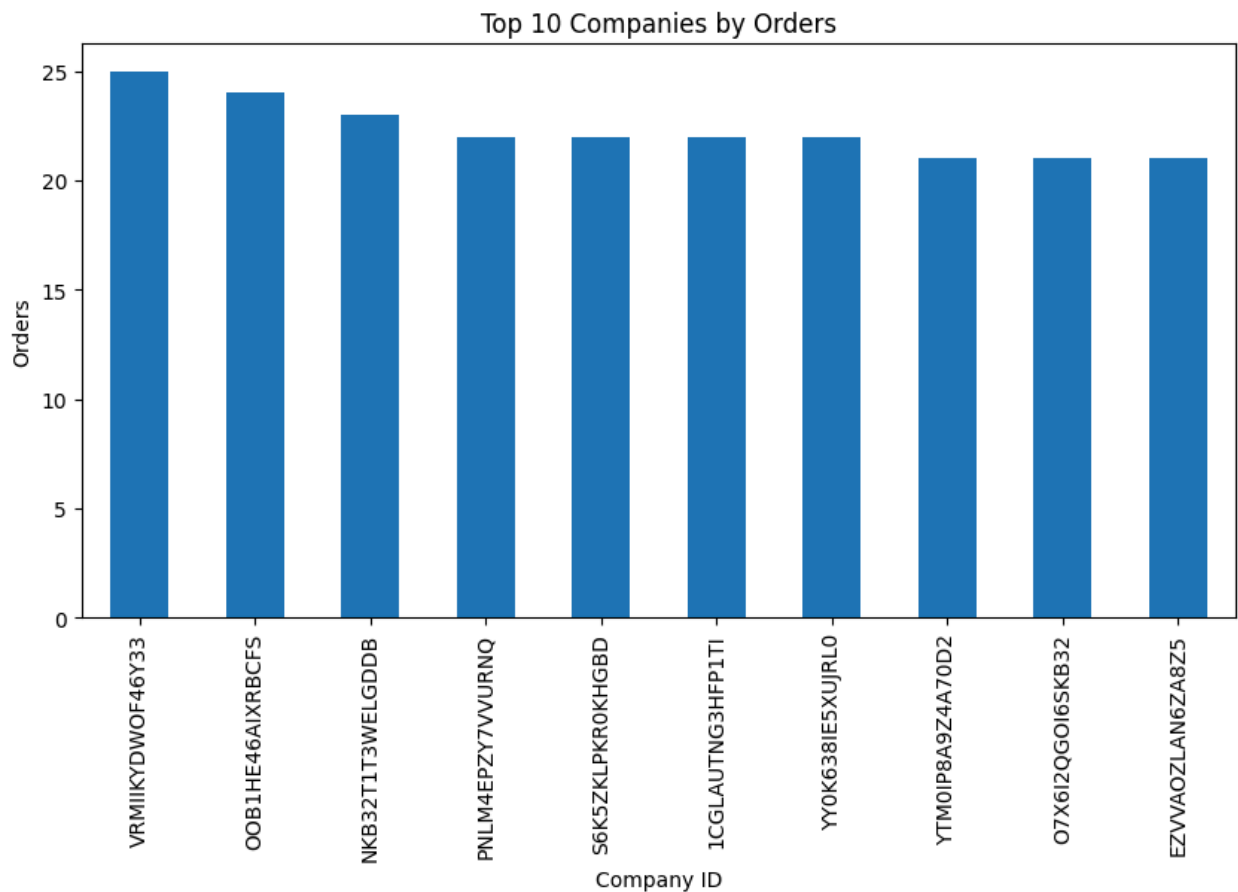
plt.figure(figsize=(8,5))
df['Year'].value_counts().sort_index().plot(kind='line', marker='o')
plt.title("Orders Over Years")
plt.xlabel("Year")
plt.ylabel("Number of Orders")
plt.show()
```



Which are the top 10 companies with the most orders?

```
In [25]: top_companies = df['Company Id'].value_counts().head(10)

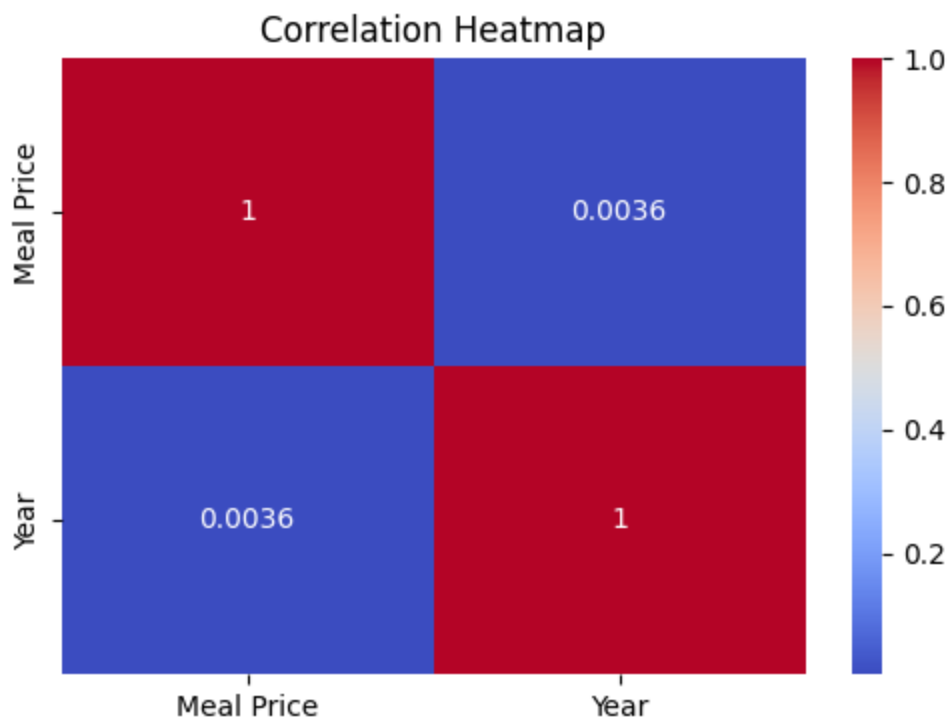
plt.figure(figsize=(10,5))
top_companies.plot(kind='bar')
plt.title("Top 10 Companies by Orders")
plt.xlabel("Company ID")
plt.ylabel("Orders")
plt.show()
```



What does the correlation heatmap show?

```
In [26]: numeric_df = df.select_dtypes(include=np.number)

plt.figure(figsize=(6,4))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



OrderLeads.csv

```
In [27]: df = pd.read_csv('/content/OrderLeads.csv')
```

How many rows and columns are in the dataset?

```
In [30]: df.shape
```

```
Out[30]: (100000, 6)
```

What are the column names in the dataset?

```
In [31]: df.columns
```

```
Out[31]: Index(['Order Id', 'Company Id', 'Company Name', 'Date', 'Order Value',  
              'Converted'],  
              dtype='object')
```

What do the first 5 records look like?

```
In [32]: df.head()
```

Out[32]:

	Order Id	Company Id	Company Name	Date	Order Value	Convert
0	80EYLOKP9E762WKG	LJKS5NK6788CYMUU	Chimera-Chasing Casbah	18-02-2017	4875	
1	TLEXR1HZWTUTBHPB	LJKS5NK6788CYMUU	Chimera-Chasing Casbah	30-07-2015	8425	
2	839FKFW2LLX4LMBB	LJKS5NK6788CYMUU	Chimera-Chasing Casbah	27-05-2016	4837	
3	97OX39BGVMHODLJM	LJKS5NK6788CYMUU	Chimera-Chasing Casbah	27-09-2018	343	
4	5T4LGH4XGBWOD49Z	LJKS5NK6788CYMUU	Chimera-Chasing Casbah	14-01-2016	983	

What do the last 5 rows look like?

In [33]: `df.tail()`

Out[33]:

	Order Id	Company Id	Company Name	Date	Order Value	Cc
99995	4MFHIAQKH0HCACJM	E4K99D4JR9E40VE1	Fierce Productivity	25-07-2015	4851	
99996	K1UP23P8SIY1YION	E4K99D4JR9E40VE1	Fierce Productivity	24-08-2015	7868	
99997	NKHFWT5I2J9LPAPG	E4K99D4JR9E40VE1	Fierce Productivity	06-09-2017	5138	
99998	OF79M3H9C44UM6PO	E4K99D4JR9E40VE1	Fierce Productivity	29-10-2015	2036	
99999	EHN2QR78K2RX0FQ9	E4K99D4JR9E40VE1	Fierce Productivity	07-03-2015	7183	

What is the overall structure of the dataset?

In [36]: `df.info()`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Order Id        100000 non-null  object
1   Company Id      100000 non-null  object
2   Company Name    100000 non-null  object
3   Date            100000 non-null  object
4   Order Value     100000 non-null  int64
5   Converted       100000 non-null  int64
dtypes: int64(2), object(4)
memory usage: 4.6+ MB

```

What statistical summary does the dataset provide?

```
In [35]: df.describe()
```

```
Out[35]:
```

	Order Value	Converted
count	100000.000000	100000.000000
mean	5051.859850	0.170290
std	2859.400416	0.375889
min	100.000000	0.000000
25%	2586.000000	0.000000
50%	5056.000000	0.000000
75%	7528.000000	0.000000
max	9999.000000	1.000000

How many missing values are present?

```
In [34]: df.isnull().sum()
```

Out[34]:

	0
Order Id	0
Company Id	0
Company Name	0
Date	0
Order Value	0
Converted	0

dtype: int64

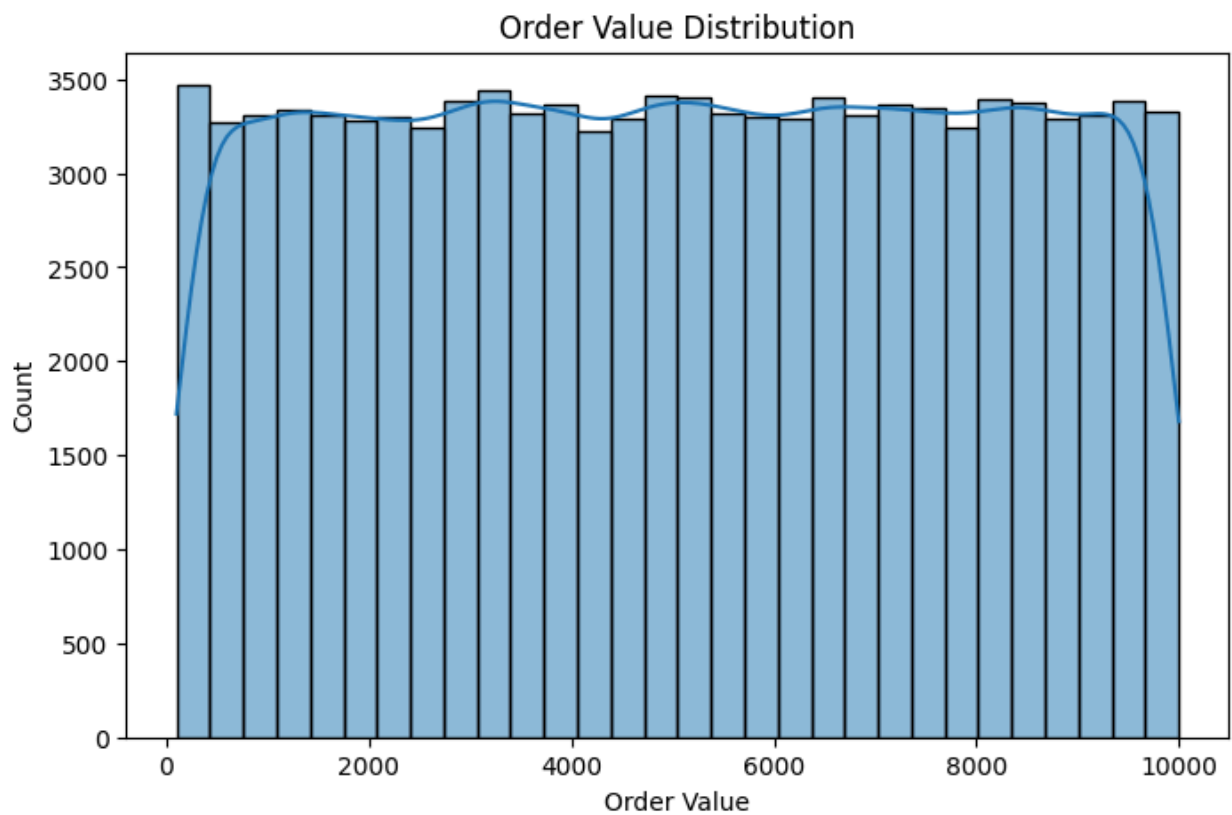
```
In [37]: df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)
```

```
In [38]: display(df.describe())
```

	Date	Order Value	Converted
count	100000	100000.000000	100000.000000
mean	2016-07-03 06:35:45.312000	5051.859850	0.170290
min	2014-01-01 00:00:00	100.000000	0.000000
25%	2015-04-02 00:00:00	2586.000000	0.000000
50%	2016-07-04 00:00:00	5056.000000	0.000000
75%	2017-10-05 00:00:00	7528.000000	0.000000
max	2018-12-31 00:00:00	9999.000000	1.000000
std	NaN	2859.400416	0.375889

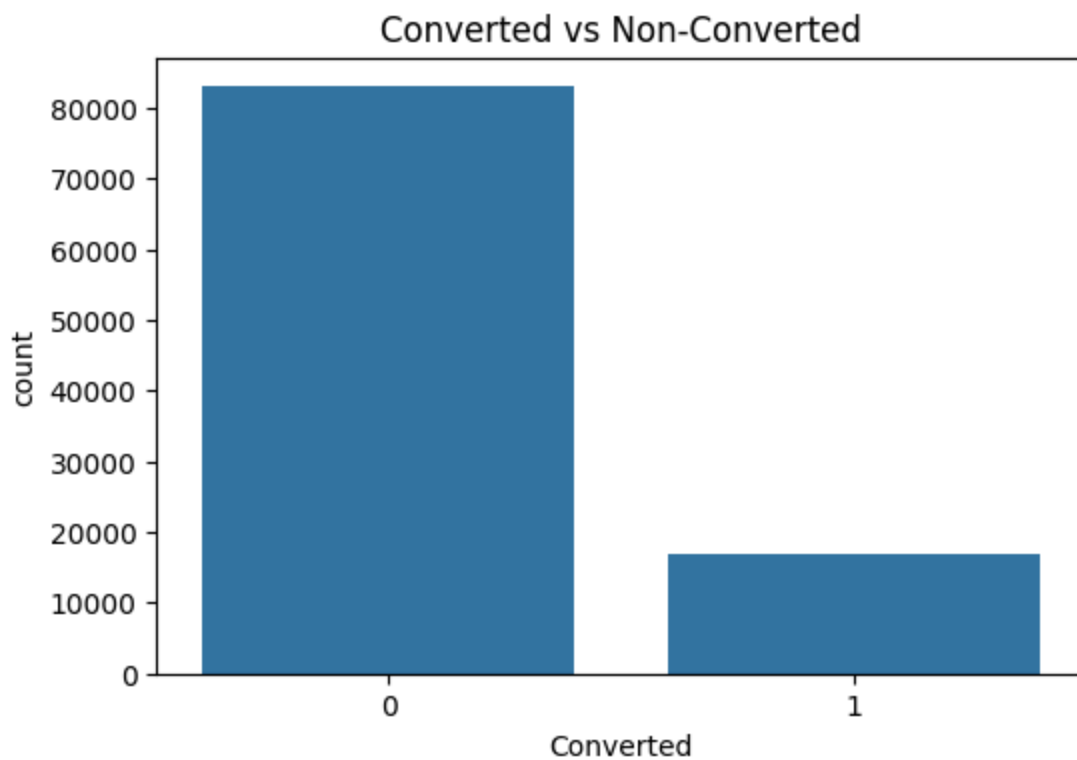
What is the distribution of Order Value?

```
In [39]: plt.figure(figsize=(8,5))
sns.histplot(df['Order Value'], bins=30, kde=True)
plt.title("Order Value Distribution")
plt.show()
```



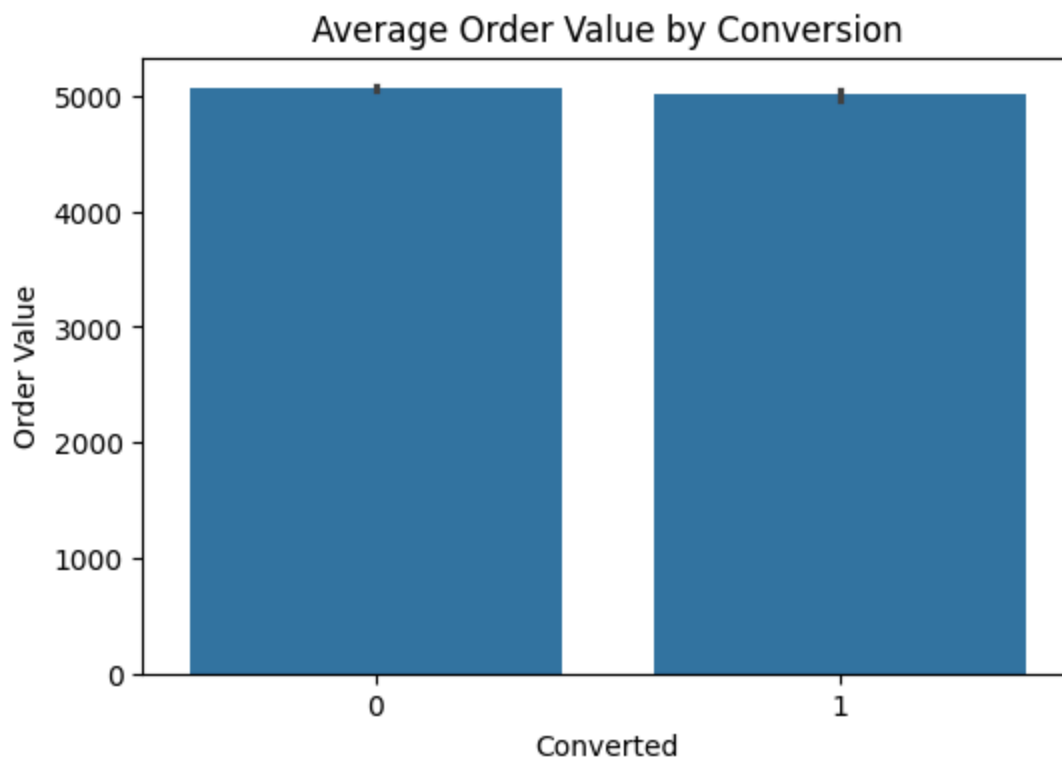
What is the ratio of converted and non-converted orders?

```
In [40]: plt.figure(figsize=(6,4))
sns.countplot(x='Converted', data=df)
plt.title("Converted vs Non-Converted")
plt.show()
```

What is the average order value of converted orders?

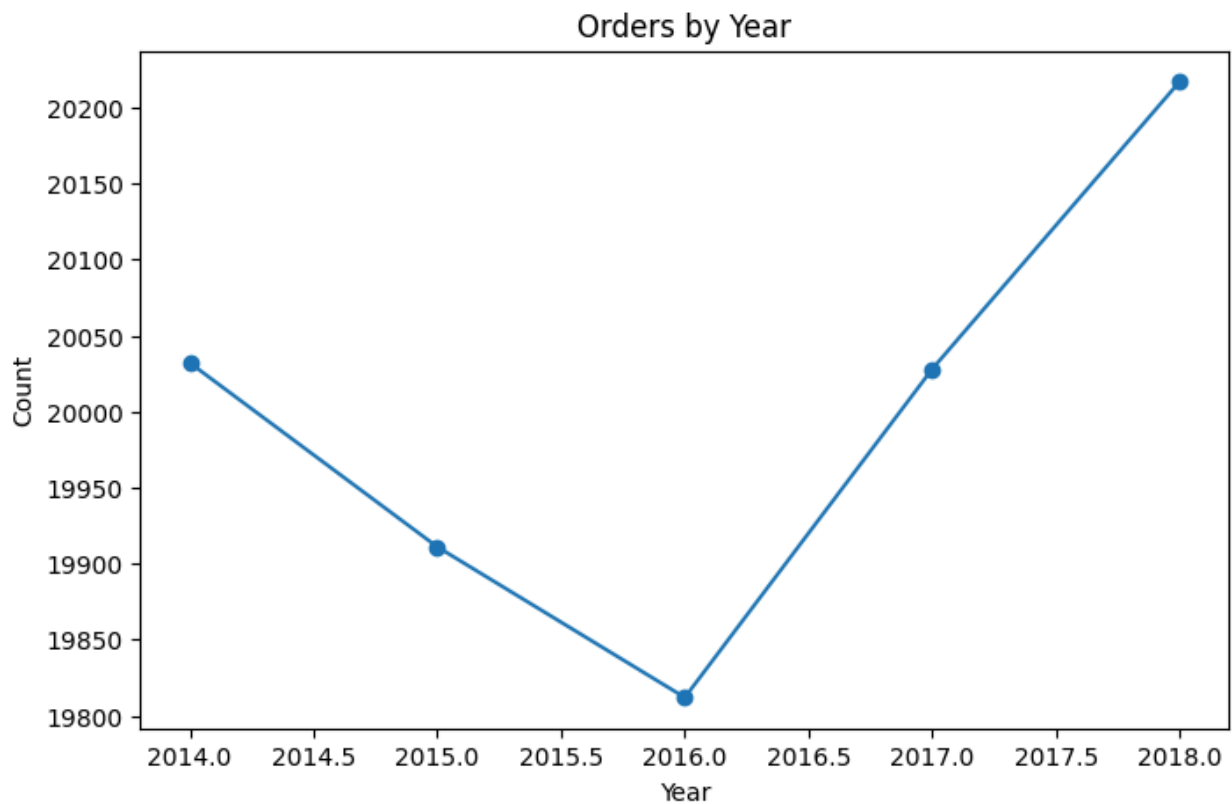
```
In [41]: plt.figure(figsize=(6,4))
sns.barplot(x='Converted', y='Order Value', data=df)
plt.title("Average Order Value by Conversion")
plt.show()
```



How have orders changed over the years?

```
In [42]: df['Year'] = df['Date'].dt.year

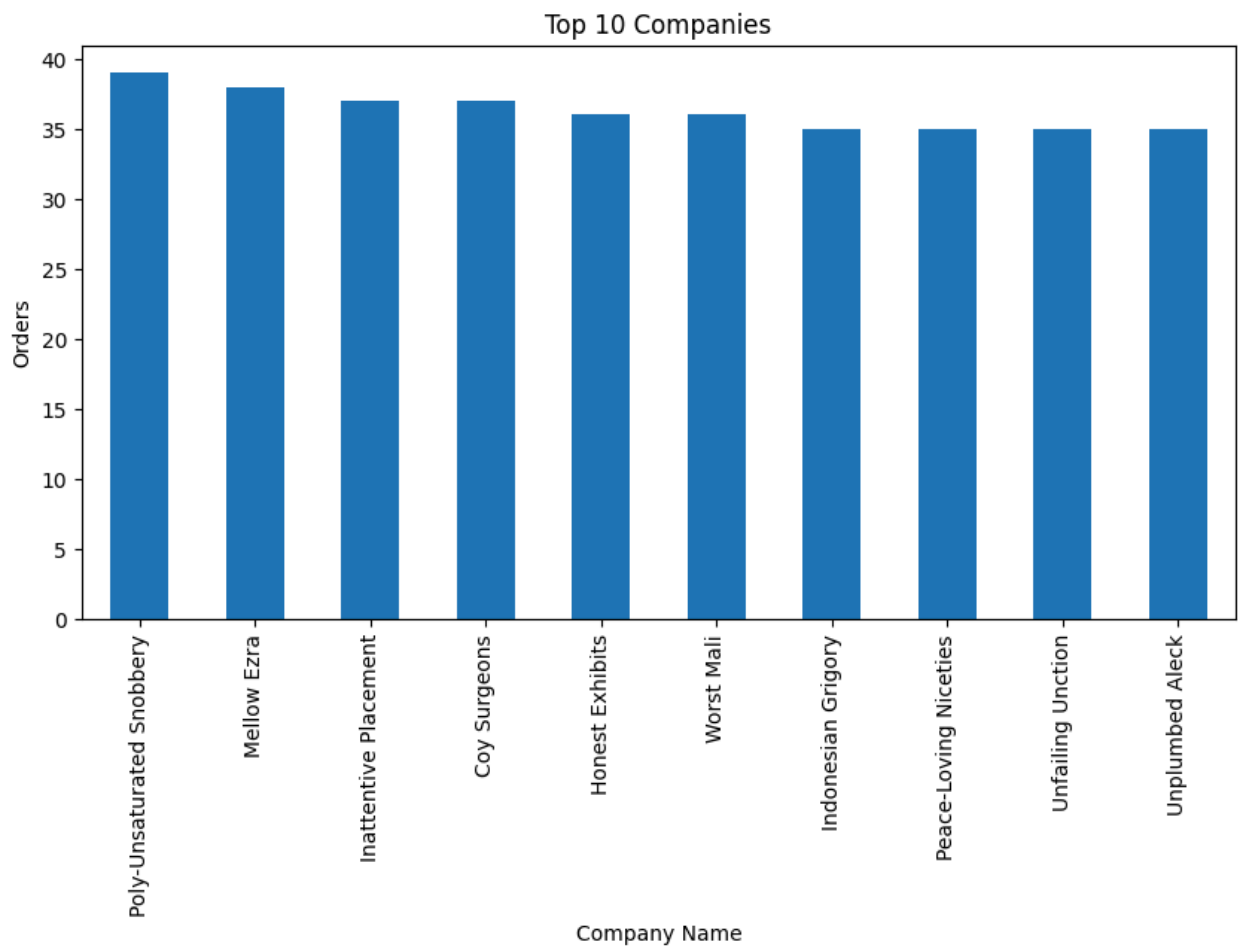
plt.figure(figsize=(8,5))
df['Year'].value_counts().sort_index().plot(kind='line', marker='o')
plt.title("Orders by Year")
plt.xlabel("Year")
plt.ylabel("Count")
plt.show()
```



Which companies have the highest number of orders?

```
In [43]: top_companies = df['Company Name'].value_counts().head(10)

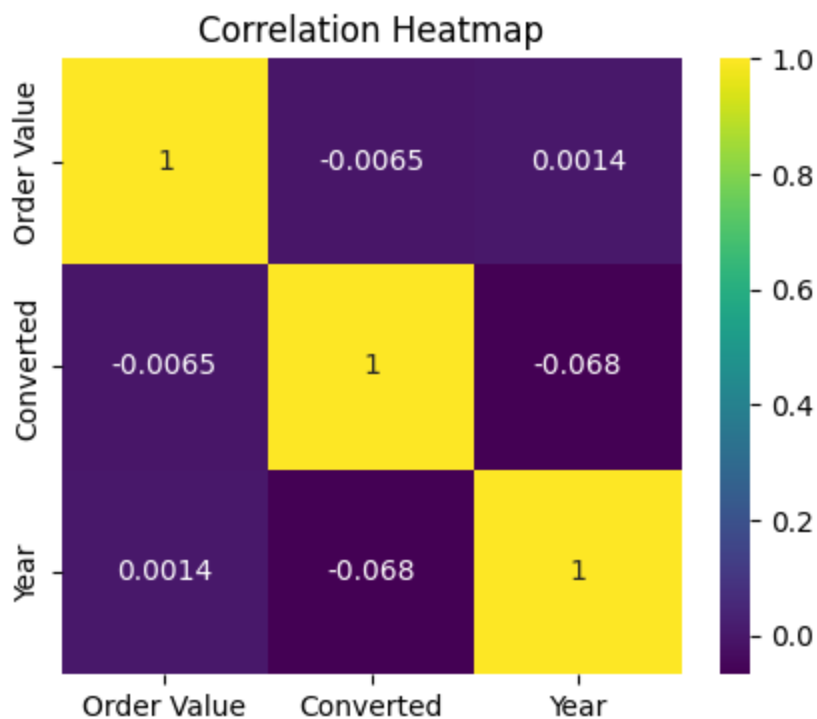
plt.figure(figsize=(10,5))
top_companies.plot(kind='bar')
plt.title("Top 10 Companies")
plt.xlabel("Company Name")
plt.ylabel("Orders")
plt.show()
```



What does the correlation heatmap show?

```
In [44]: numeric_df = df.select_dtypes(include=np.number)

plt.figure(figsize=(5,4))
sns.heatmap(numeric_df.corr(), annot=True, cmap='viridis')
plt.title("Correlation Heatmap")
plt.show()
```



SalesTeam.csv

```
In [45]: df = pd.read_csv('/content/SalesTeam.csv')
```

How many rows and columns are in the dataset?

```
In [46]: df.shape
```

```
Out[46]: (4725, 4)
```

What are the column names in the dataset?

```
In [47]: df.columns
```

```
Out[47]: Index(['Sales Rep', 'Sales Rep Id', 'Company Name', 'Company Id'], dtype='object')
```

What do the first 5 rows look like?

```
In [48]: df.head()
```

Out[48]:

	Sales Rep	Sales Rep Id	Company Name	Company Id
0	Jessie Mcallister	97UNNAT790E0WM4N	Chimera-Chasing Casbah	LJKS5NK6788CYMUU
1	Jessie Mcallister	97UNNAT790E0WM4N	Tangential Sheds	36MFTZOYMTAJP1RK
2	Jessie Mcallister	97UNNAT790E0WM4N	Two-Mile Grab	H3JRC7XX7WJAD4ZO
3	Jessie Mcallister	97UNNAT790E0WM4N	Three-Men-And-A-Helper Congo'S	HB25MDZR0MGCQUGX
4	Jessie Mcallister	97UNNAT790E0WM4N	Biophysical Battleground	7RVA8TIVBLBXMNO4

What do the last 5 rows look like?

In [49]: `df.tail()`

Out[49]:

	Sales Rep	Sales Rep Id	Company Name	Company Id
4720	Joel Woodbury	YBWRC1WQHXE895WV	Jammed-Together Stable	4JW3GG4YW52M86F5
4721	Joel Woodbury	YBWRC1WQHXE895WV	Laureate Barbitol	R0KV9HJ7RIRRYSVG
4722	Hazel Dickerson	NIW73ZQMGEOB7BHT	Spotty Adoniram	5K17HYJQ6JG6UFU0
4723	Hazel Dickerson	NIW73ZQMGEOB7BHT	Physicochemical Impatience	FC1LJXHNH2883N3Y
4724	Hazel Dickerson	NIW73ZQMGEOB7BHT	Fierce Productivity	E4K99D4JR9E40VE1

What is the overall structure of the dataset?

In [50]: `df.info()`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4725 entries, 0 to 4724
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Sales Rep       4725 non-null   object
1   Sales Rep Id    4725 non-null   object
2   Company Name    4725 non-null   object
3   Company Id      4725 non-null   object
dtypes: object(4)
memory usage: 147.8+ KB

```

What statistical summary does the dataset provide?

```
In [51]: df.describe()
```

```
Out[51]:
```

	Sales Rep	Sales Rep Id	Company Name	Company Id
count	4725	4725	4725	4725
unique	500	500	4725	4725
top	Oscar Mantini	ISA8445OKV13F3YZ	Fierce Productivity	E4K99D4JR9E40VE1
freq	18	18	1	1

How many missing values are present?

```
In [52]: df.isnull().sum()
```

```
Out[52]:
```

	0
Sales Rep	0
Sales Rep Id	0
Company Name	0
Company Id	0

dtype: int64

How many duplicate rows are present?

```
In [53]: print("\nDuplicates:", df.duplicated().sum())
```

Duplicates: 0

How many unique sales representatives are there?

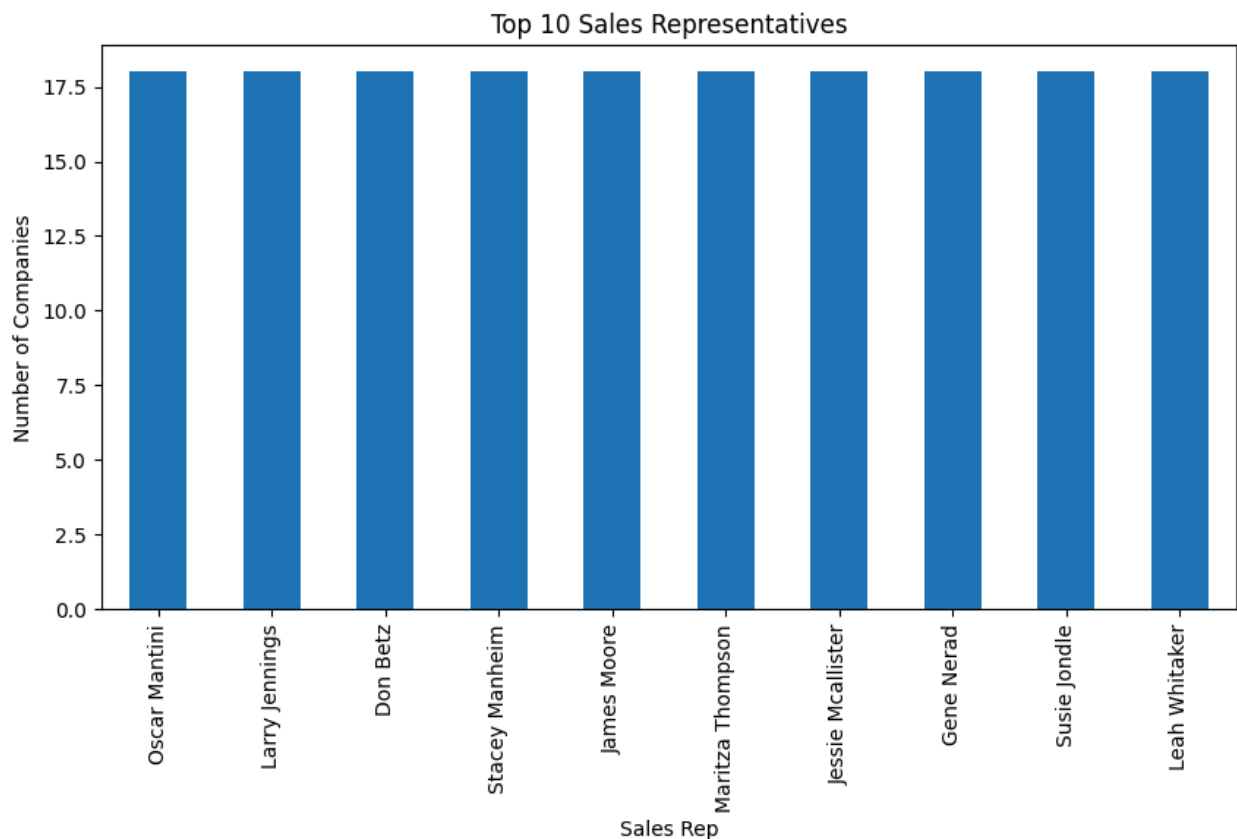
```
In [56]: print("\nUnique Sales Reps:", df['Sales Rep'].nunique())
print("Unique Companies:", df['Company Name'].nunique())
```

Unique Sales Reps: 500
Unique Companies: 4725

Which sales representatives manage the highest number of companies?

```
In [57]: top_reps = df['Sales Rep'].value_counts().head(10)

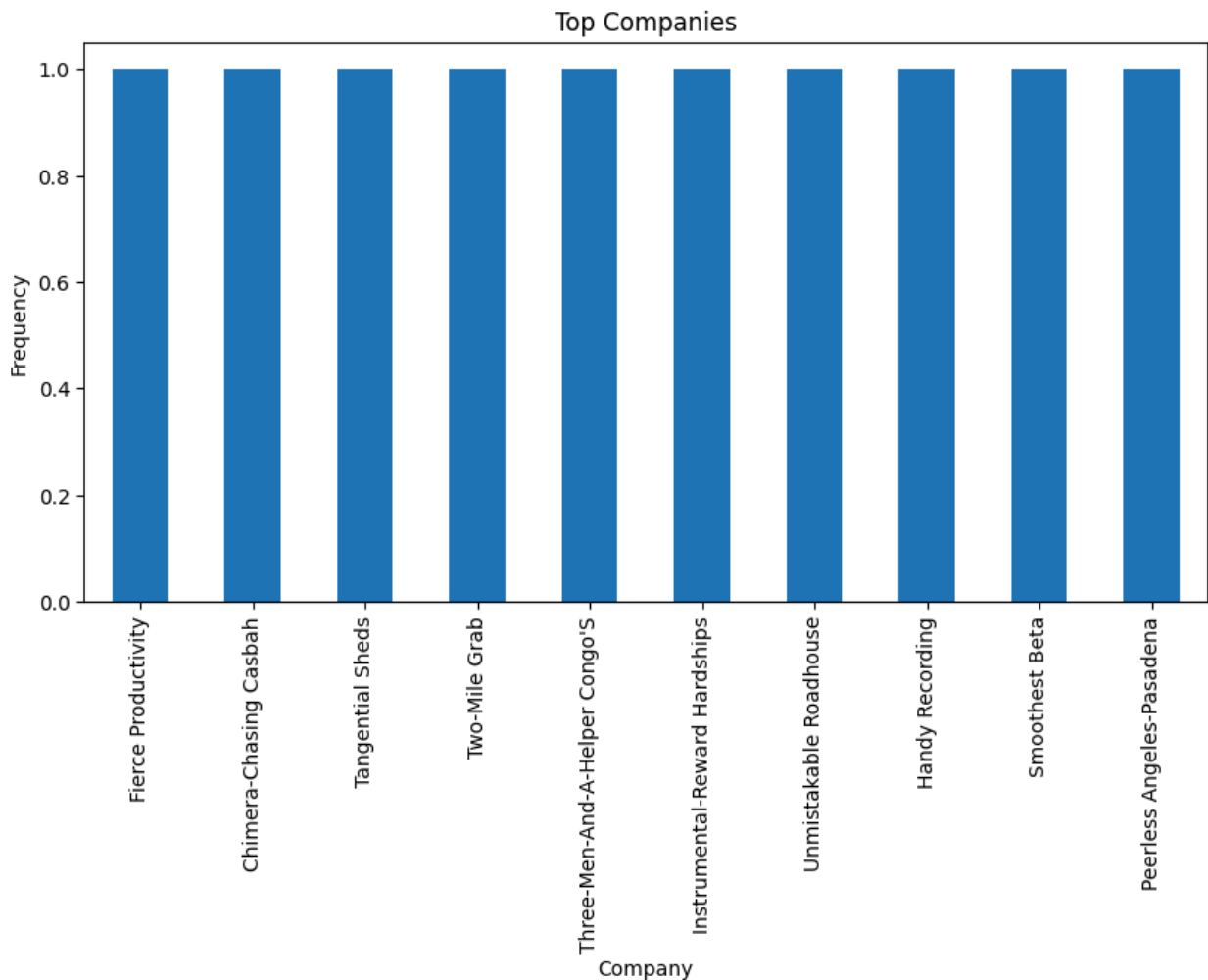
plt.figure(figsize=(10,5))
top_reps.plot(kind='bar')
plt.title("Top 10 Sales Representatives")
plt.xlabel("Sales Rep")
plt.ylabel("Number of Companies")
plt.show()
```



Which companies appear most frequently in the dataset?

```
In [58]: top_companies = df['Company Name'].value_counts().head(10)

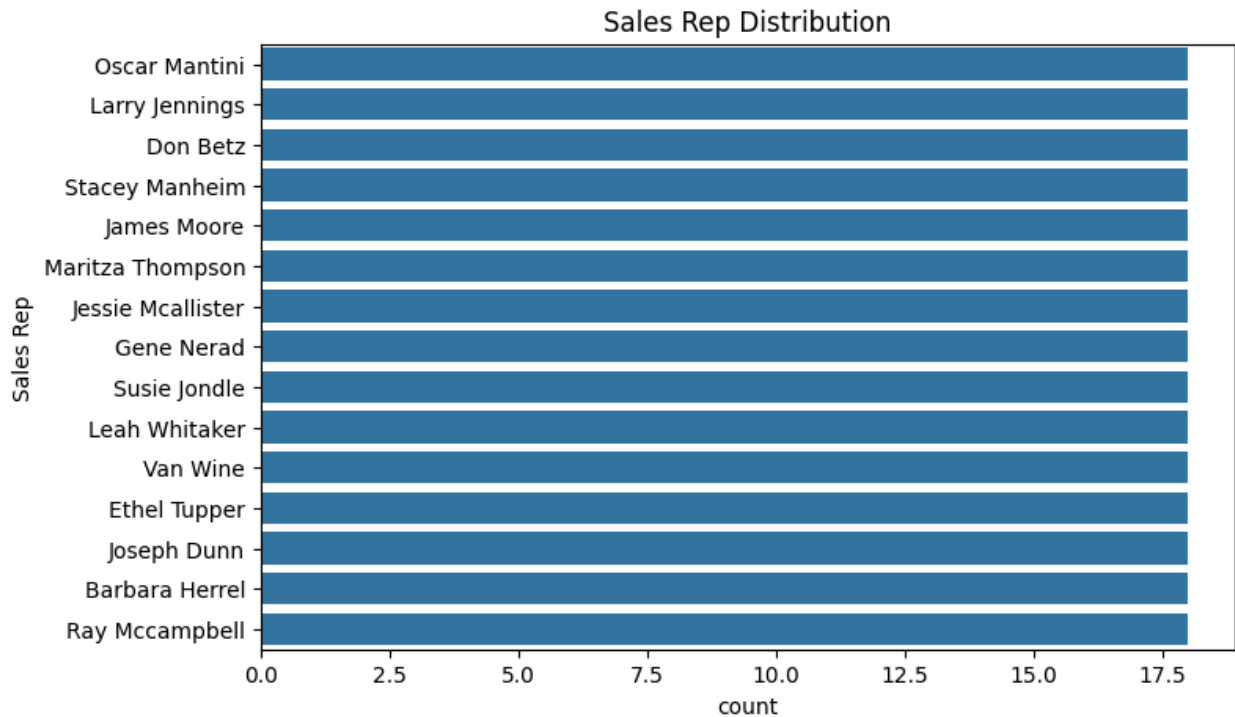
plt.figure(figsize=(10,5))
top_companies.plot(kind='bar')
plt.title("Top Companies")
plt.xlabel("Company")
plt.ylabel("Frequency")
plt.show()
```



What is the distribution of sales representatives in the dataset?

```
In [59]: plt.figure(figsize=(8,5))
sns.countplot(y='Sales Rep', data=df,
              order=df['Sales Rep'].value_counts().head(15).index)
plt.title("Sales Rep Distribution")
```

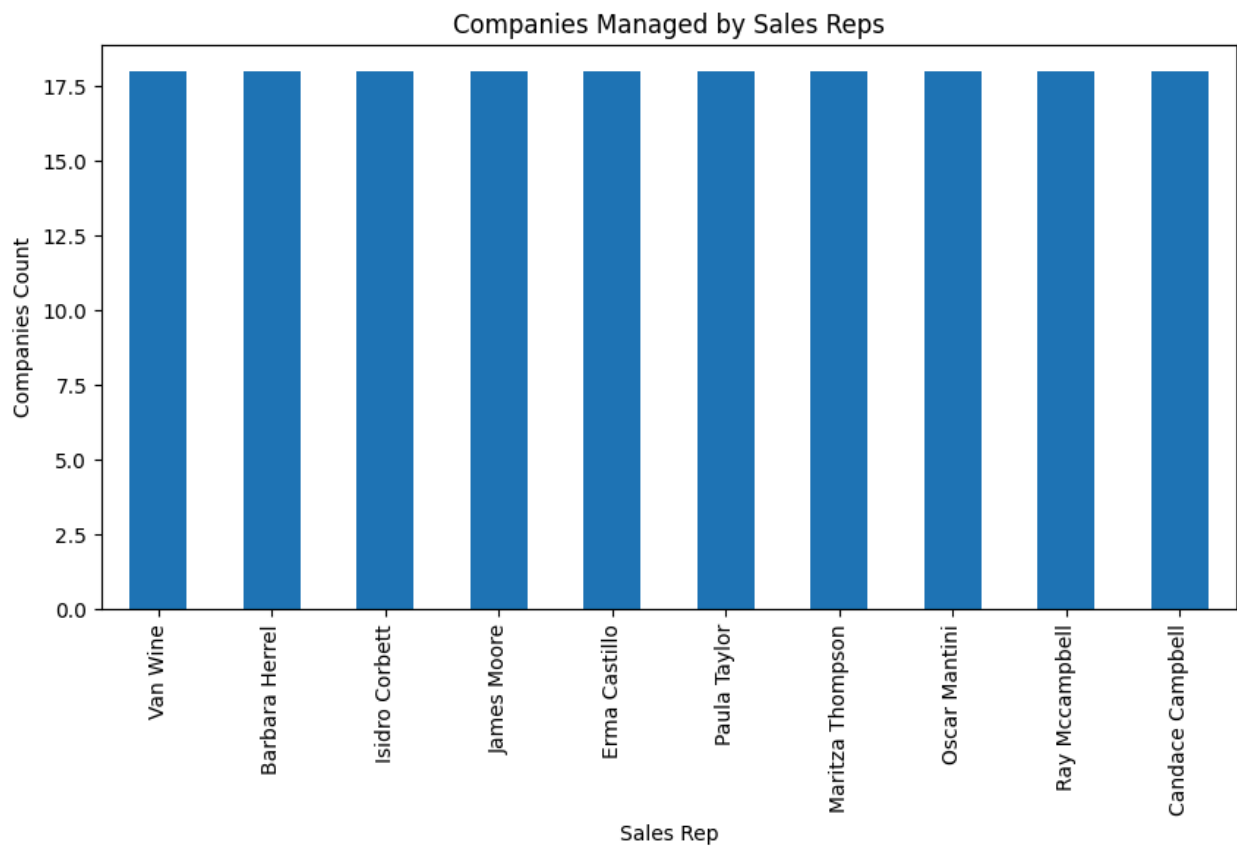
```
plt.show()
```



How many companies are managed by each sales representative?

```
In [60]: company_per_rep = df.groupby('Sales Rep')['Company Id'].count().sort_values(ascending=True)

plt.figure(figsize=(10,5))
company_per_rep.head(10).plot(kind='bar')
plt.title("Companies Managed by Sales Reps")
plt.xlabel("Sales Rep")
plt.ylabel("Companies Count")
plt.show()
```



Conclusion

The complete EDA project successfully provided valuable insights into customer behavior, sales performance, operational trends, and workforce management. By analyzing the Invoices, OrderLeads, and SalesTeam datasets together, it became possible to understand multiple aspects of business operations through data-driven analysis.

Several important findings were identified during the project:

- Certain companies consistently generated higher order activity
- Some meal categories were significantly more popular than others
- Converted orders generally had higher order values
- Order trends changed over time, showing business growth patterns
- Sales team workload distribution was uneven across representatives
- Correlation analysis revealed meaningful relationships between business variables

Multiple visualizations were created throughout the analysis, including:

- Histograms

- Bar charts
- Count plots
- Line charts
- Correlation heatmaps

These visualizations helped make complex business data easier to understand and interpret.

The project also demonstrated the importance of data cleaning, preprocessing, and exploratory analysis in real-world business environments. Through this analysis, several business policies and recommendations were identified, including:

- Improving customer retention strategies
- Optimizing pricing and inventory planning
- Focusing on high-converting leads
- Balancing workload among sales representatives
- Implementing performance monitoring systems
- Using forecasting techniques for future planning

Overall, this project highlights how Exploratory Data Analysis can transform raw datasets into meaningful business intelligence. The analysis supports better decision-making, improves operational efficiency, and helps organizations make strategic business decisions using real data insights.